

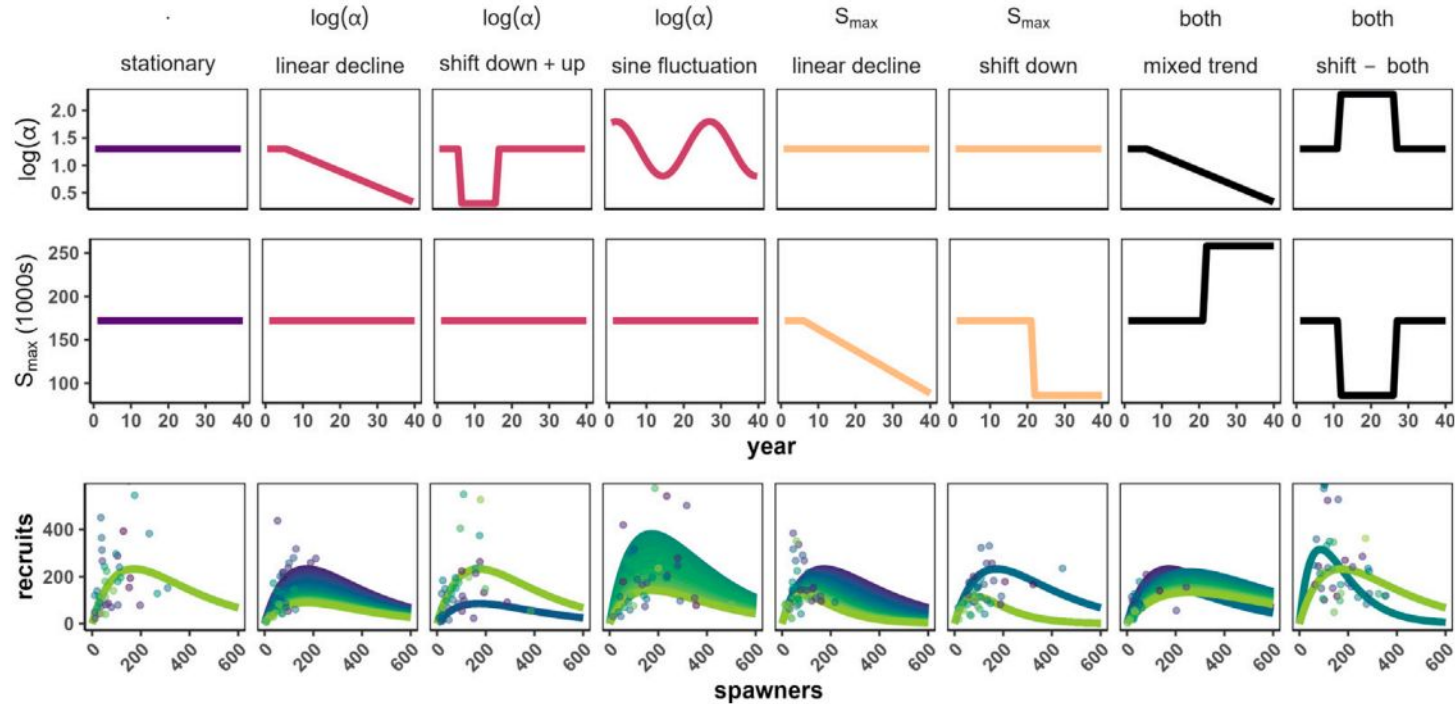
Nonstationary spawner-recruitment dynamics for Pacific salmon workshop



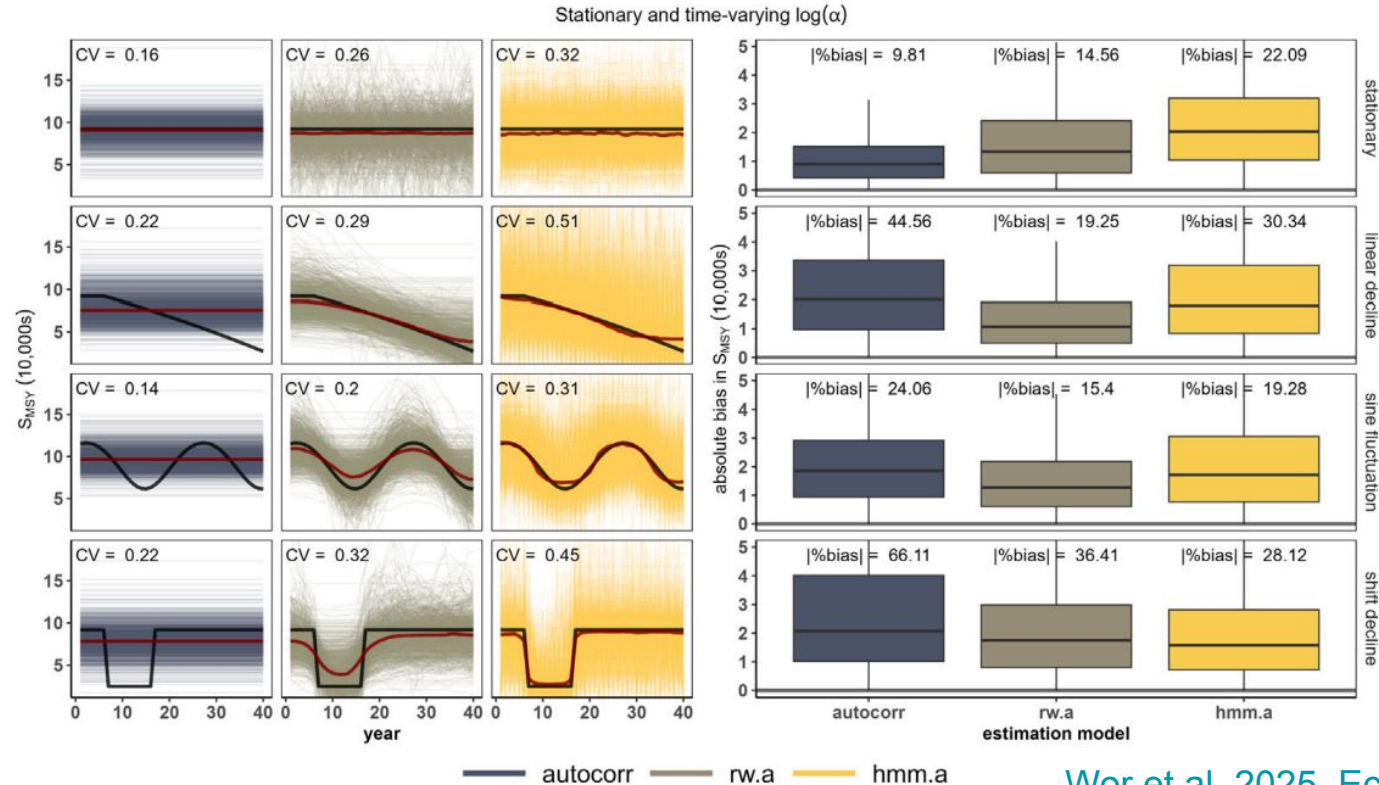
**Dan Greenberg
Dylan Glaser**

**Salmon Assessment Section
Pacific Biological Station
Fisheries and Oceans Canada**

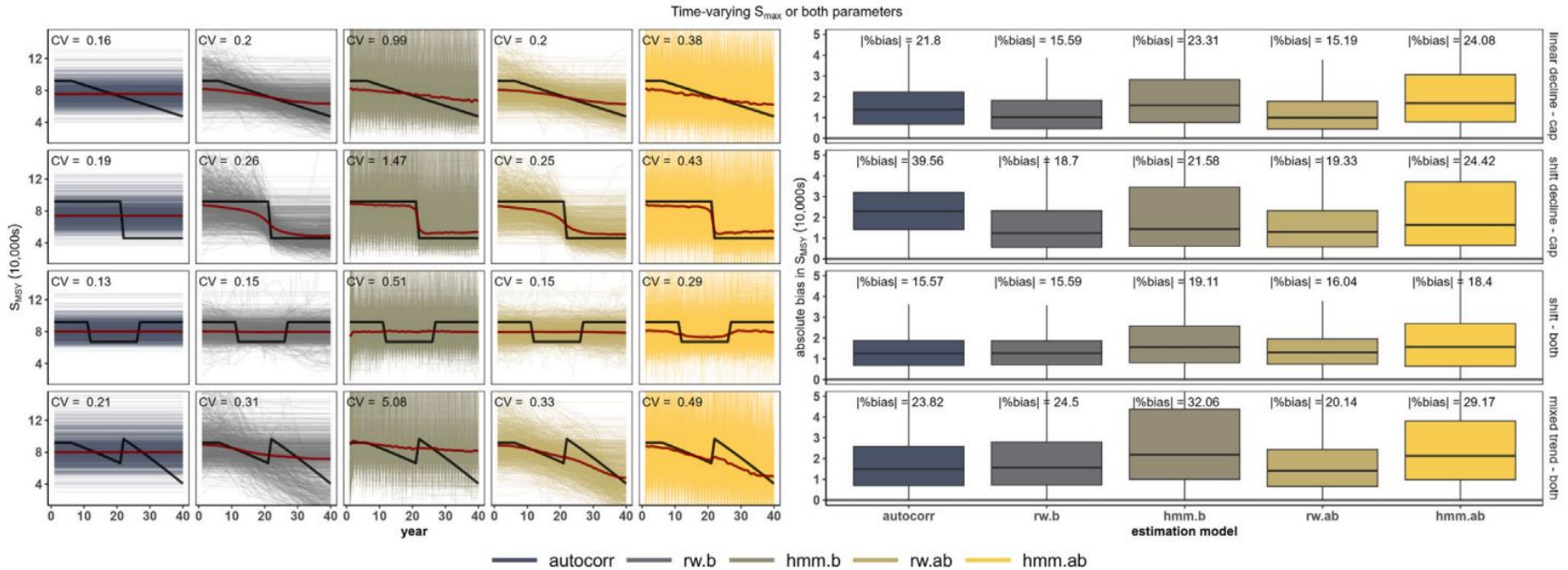
Non-stationary models work - *on average* *



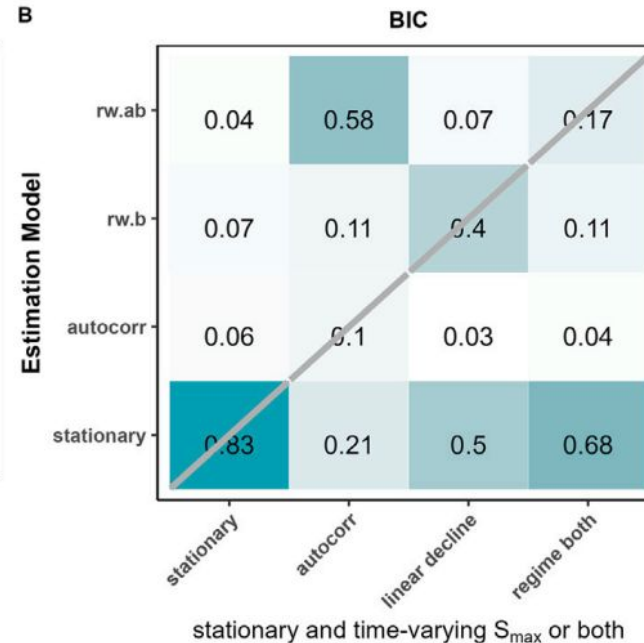
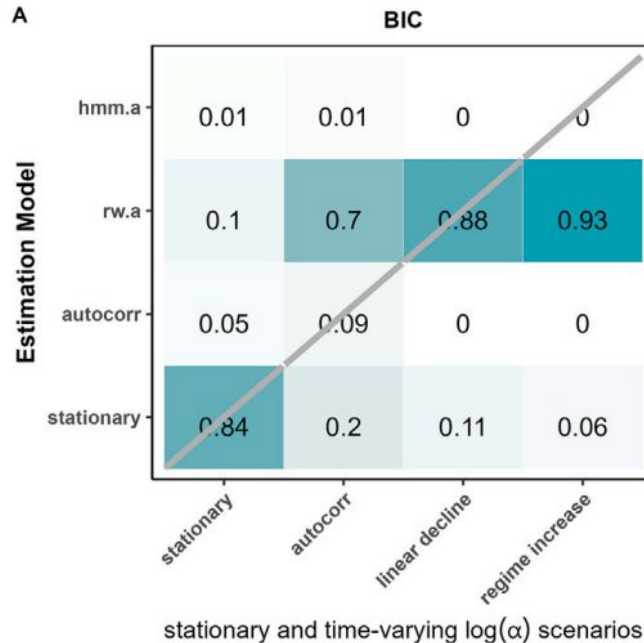
Non-stationary models work - *on average* *



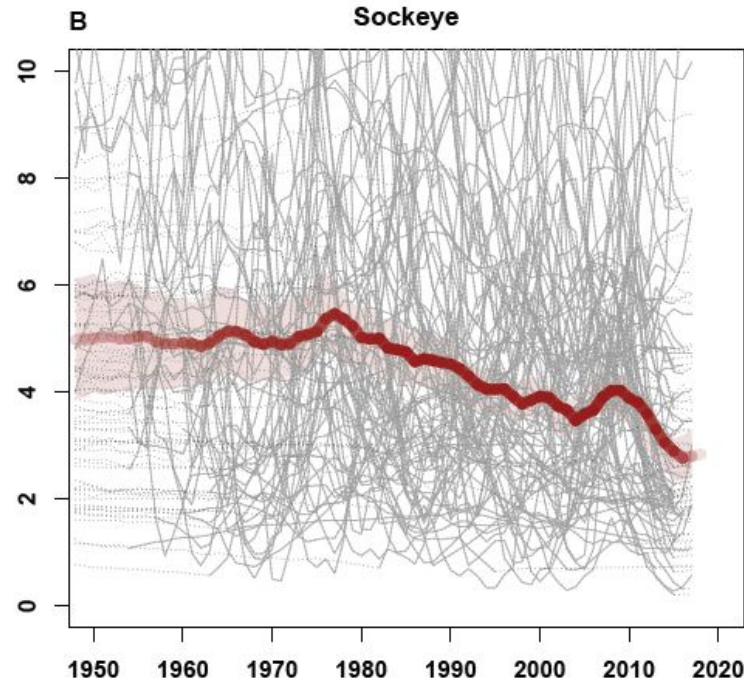
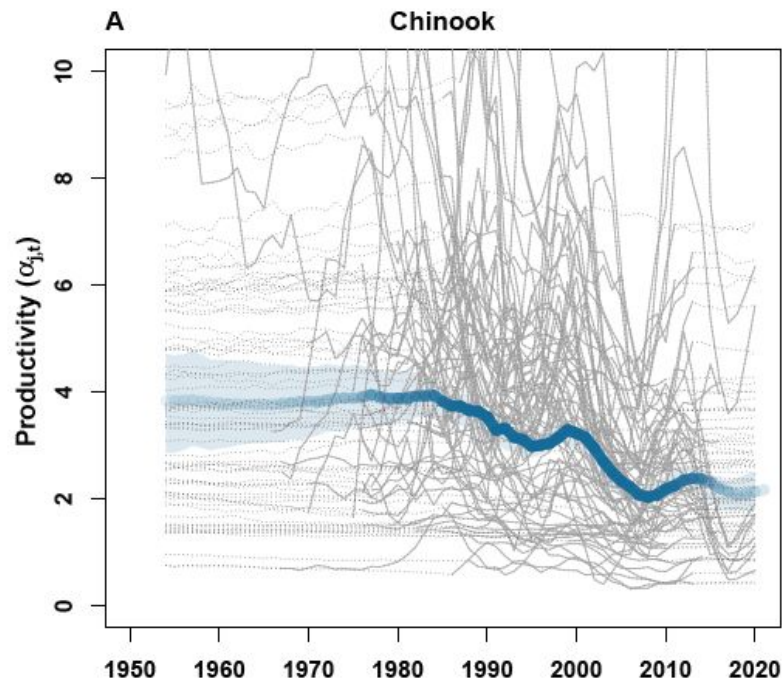
Non-stationary models work - *on average* *



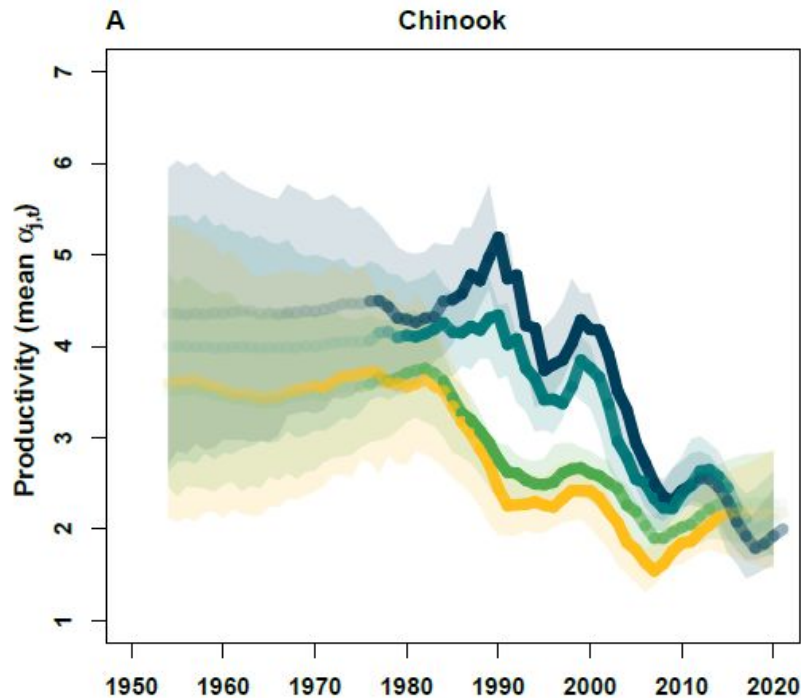
Model selection often misses the correct generating model



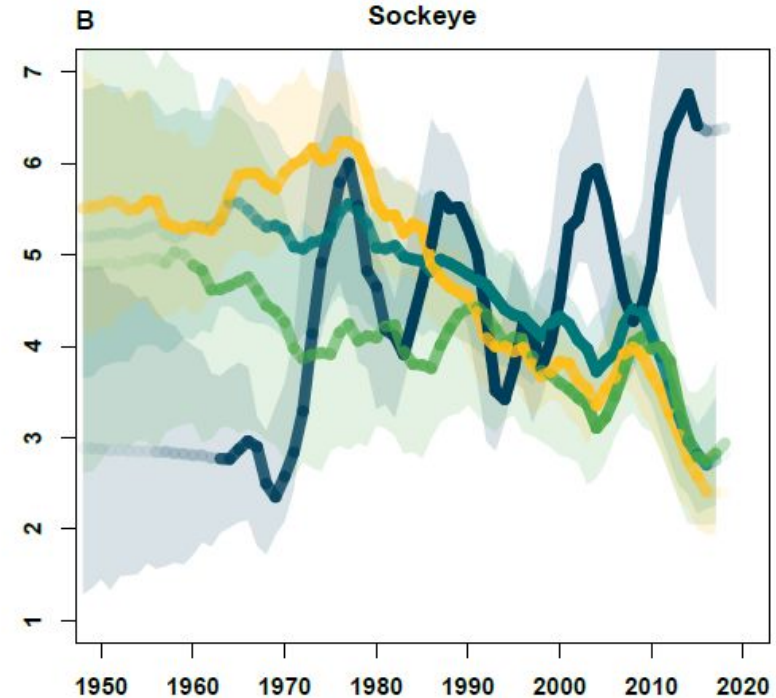
Empirical evidence suggests ubiquitous productivity changes - but are they real?



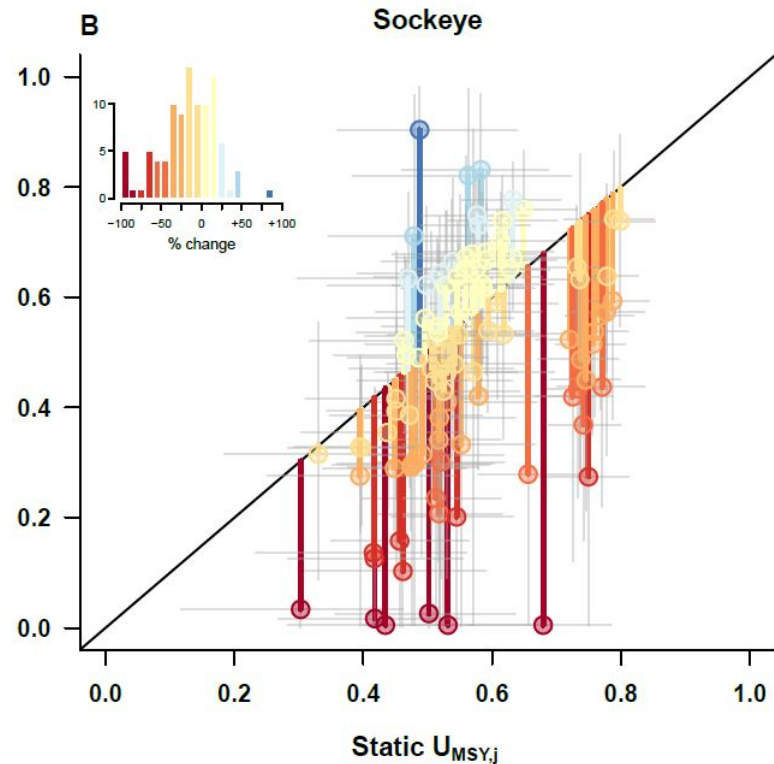
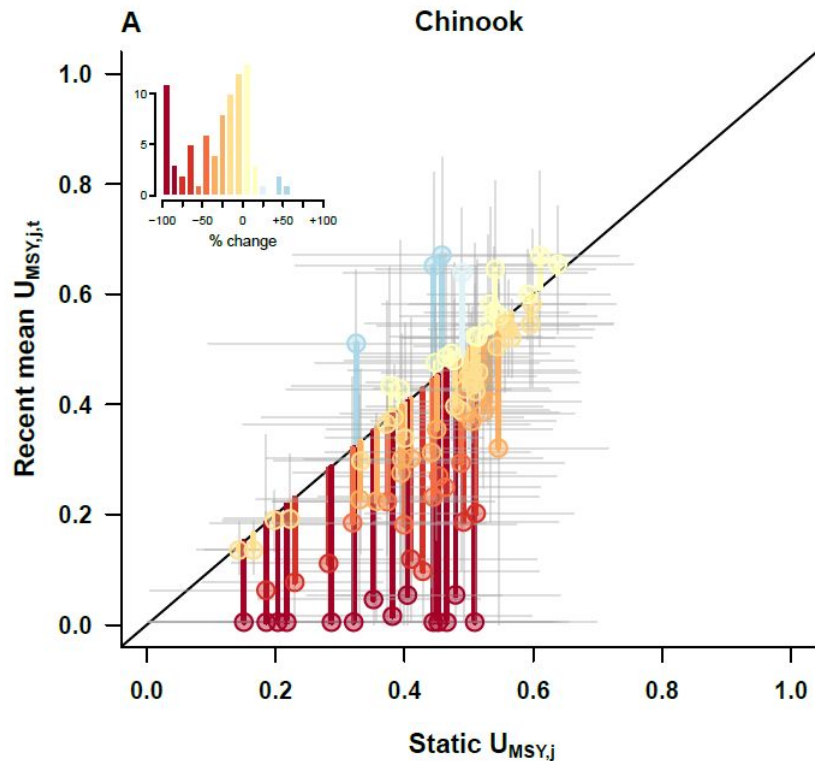
Patterns of nonstationarity can be both similar or very different across regions



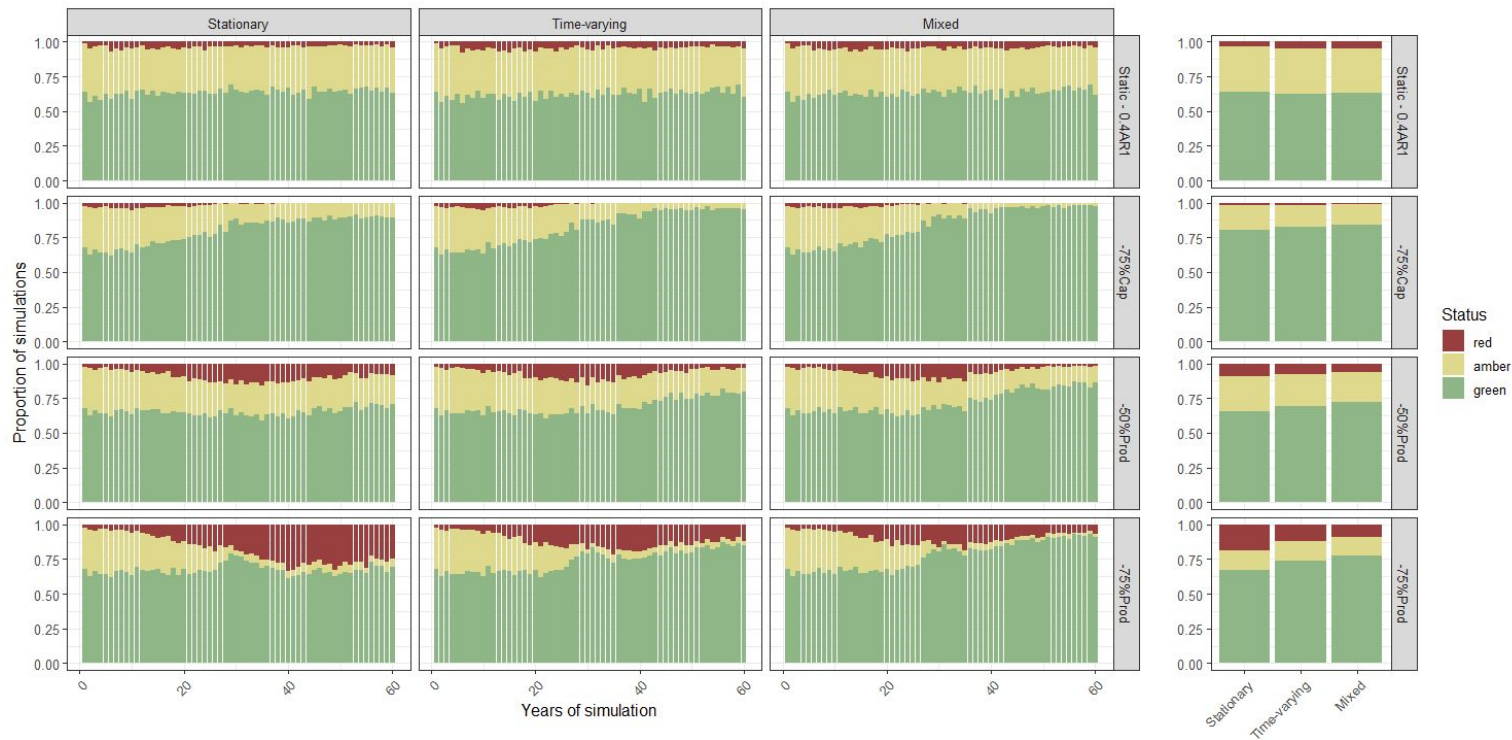
South Coast North Coast Gulf of Alaska Bering Sea



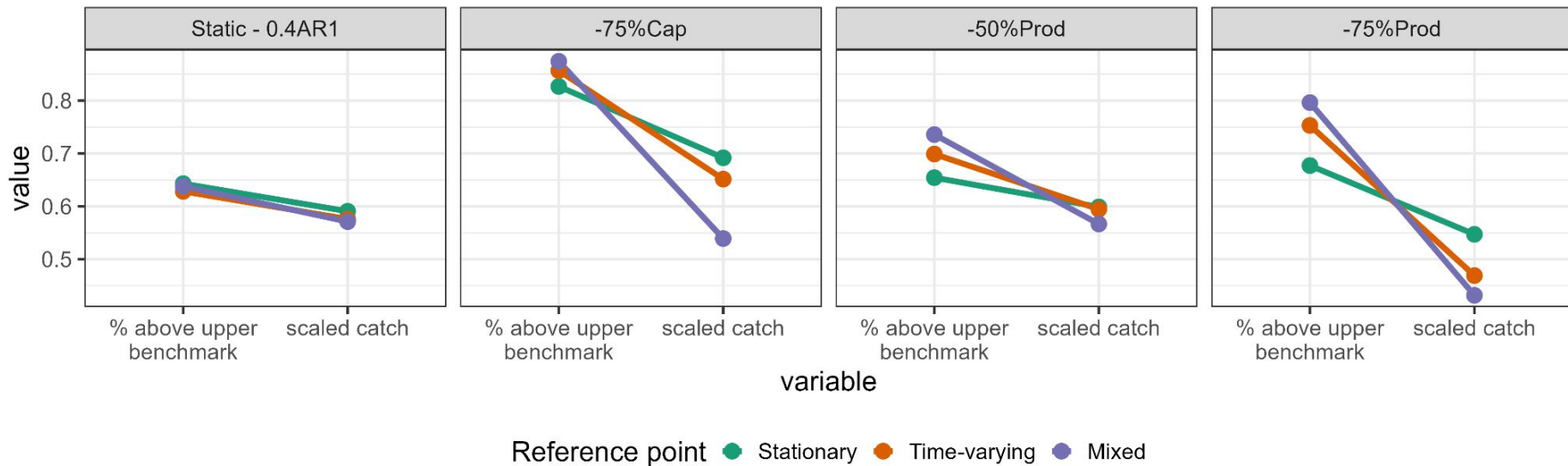
These changes may have considerable consequences for sustainable management



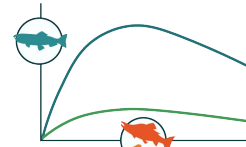
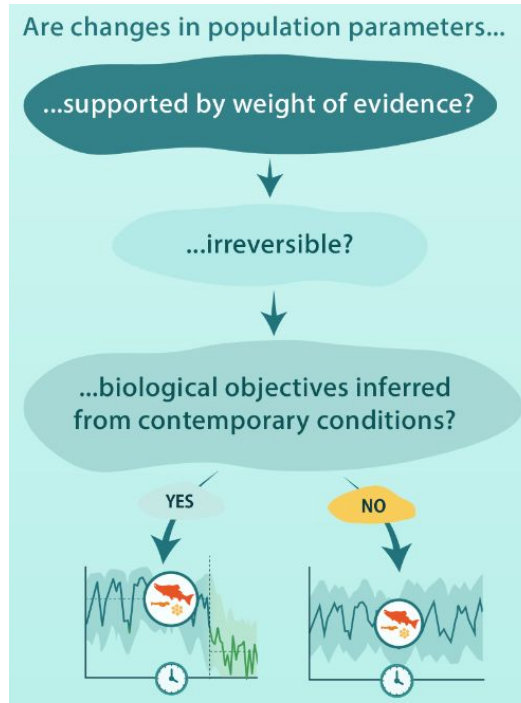
Non-stationary reference points may have advantages, under certain contexts



But these come with trade-offs and can be very context specific



Given all this... to what degree should nonstationarity be considered in stock assessment?



Spawner-recruitment relationships



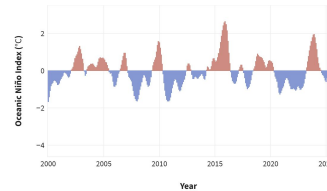
Size/growth



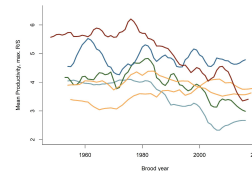
Fecundity



Maturity/age structure



Ecosystem indicators (abiotic and biotic)



Meta-analyses

Given all this... to what degree should nonstationarity be considered in management?

If nonstationary dynamics seem to be supported:

Test alternative scenarios (in terms of baseline/future productivity/capacity and their incorporation into different harvest control rules) in a management strategy evaluation - tailored to system-specific dynamics

